

EQ Credit: Construction Indoor Air Quality Management Plan

This credit applies to

- ▶ BD+C: New Construction (1 point)
- ▶ BD+C: Core & Shell (1 point)
- ▶ BD+C: Schools (1 point)
- ▶ BD+C: Retail (1 point)
- ▶ BD+C: Data Centers (1 point)
- ▶ BD+C: Warehouses & Distribution Centers (1 point)
- ▶ BD+C: Hospitality (1 point)
- ▶ BD+C: Healthcare (1 point)

INTENT

To promote the well-being of construction workers and building occupants by minimizing indoor air quality problems associated with construction and renovation.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY

Develop and implement an indoor air quality (IAQ) management plan for the construction and preoccupancy phases of the building. The plan must address all of the following.

During construction, meet or exceed all applicable recommended control measures of the Sheet Metal and Air Conditioning National Contractors Association (SMACNA) IAQ Guidelines for Occupied Buildings under Construction, 2nd edition, 2007, ANSI/SMACNA 008-2008, Chapter 3.

Protect absorptive materials stored on-site and installed from moisture damage.

Do not operate permanently installed air-handling equipment during construction unless filtration media with a minimum efficiency reporting value (MERV) of 8, as determined by ASHRAE 52.2-2017, with errata (or media with ISO_{coarse} 90% or higher, as defined by ISO 16890-2016, Particulate Air Filters for General Ventilation, Determination of the Filtration Performance), are installed at each return air grille and return or transfer duct inlet opening such that there is no bypass around the filtration media. Immediately before occupancy, replace all filtration media with the final design filtration media, installed in accordance with the manufacturer's recommendations.

Prohibit the use of smoking inside the building and within 25 feet (7.5 meters) of the building openings during construction. Smoking includes tobacco smoke, as well as smoke produced from the combustion of cannabis and controlled substances and the emissions produced by electronic smoking devices.

HEALTHCARE

Moisture. Develop and implement a moisture control plan to protect stored on-site and installed absorptive materials from moisture damage. Immediately remove from site and properly dispose of any materials susceptible to microbial growth and replace with new, undamaged materials. Also include strategies for protecting the building from moisture intrusion and preventing occupants' exposure to mold spores.

Particulates. Do not operate permanently installed air-handling equipment during construction unless filtration media with a minimum efficiency reporting value (MERV) of 8, as determined by ASHRAE 52.2-2017, with errata (or media with ISO_{coarse} 90% or higher, as defined by ISO 16890-2016, Particulate Air Filters for General Ventilation, Determination of the Filtration Performance), are installed at each return air grille and return or transfer duct inlet opening such that there is no bypass around the filtration

media. Immediately before occupancy, replace all filtration media with the final design filtration media, installed in accordance with the manufacturer's recommendations.

VOCs. Schedule construction procedures to minimize exposure of absorbent materials to VOC emissions. Complete painting and sealing before storing or installing "dry" materials, which may accumulate pollutants and release them over time. Store fuels, solvents, and other sources of VOCs separately from absorbent materials.

Outdoor emissions. For renovation projects involving waterproofing, repairing asphalt roofing, sealing parking lots, or other outdoor activities that generate high VOC emissions, develop a plan to manage fumes and avoid infiltration to occupied spaces. Comply with the procedures established by NIOSH, *Asphalt Fume Exposures during the Application of Hot Asphalt to Roofs* (Publication 2003-112).

Tobacco. Prohibit the use of tobacco products inside the building and within 25 feet (7.5 meters) of the building entrance during construction.

Noise and vibration. Develop a plan based on the British Standard (BS 5228) to reduce noise emissions and vibrations from construction equipment and other nonroad engines by specifying low-noise emission design or the lowest decibel level available that meets performance requirements in the British Standard. Construction crews must wear ear protection in areas where sound levels exceed 85 dB for extended periods.

Infection control. For renovations and additions adjacent to occupied facilities or phased occupancy in new construction, follow the FGI 2018 Guidelines for Design and Construction of Hospitals, Guidelines for Design and Construction of Outpatient Facilities, Guidelines for Design and Construction of Residential Health, Care, and Support Facilities and The Joint Commission Standards to establish an integrative infection control team comprising the owner, designer, and contractor to evaluate infection control risk and document the required precautions in a project-specific plan. Use the Guidelines for Environmental Infection Control in Health-Care Facilities, 2003, updated July 2019 to assess risk and to select mitigation procedures for construction activities.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Changes for this credit include updates to recent versions of referenced ASHRAE, FGI and European standards or updates to reference more globally applicable standards. The changes made were to the following standards:

- ▶ ASHRAE 52.2-2010 → ASHRAE 52.2-2017 (updated version)
- ▶ EN 779-2002 → ISO 16890-2016 (global applicability)
- ▶ FGI 2010 → FGI 2018

Other clarifications were including a more extensive definition of smoking to cover related non-tobacco products and to extend the prohibition of smoking within 25 feet of all building openings during construction not just entrances.

Step-by-Step Guidance

Refer to LEED v4 reference guide with the following modifications:

Healthcare

Step 1. Evaluate infection control risk: Follow the newer 2018 FGI Guidelines for Design and Construction of Health Care facilities.

Further Explanation

International Tips

In countries where MERV ratings are not available, filtration class used must be ISO_{coarse} 90% or higher per ISO 16890-2016, Particulate Air Filters for General Ventilation, Determination of the Filtration Performance instead of Class F5 per EN 779-2002.

Connection to Ongoing Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to promote well-being of construction workers and occupants during construction activities such as formulating and implementing an indoor air quality plan can contribute to better indoor environmental quality during operations phase.